

MD NASIR UDDIN

Research Associate | Software Engineer | Machine Learning & AI
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PROFESSIONAL SUMMARY

Software Engineering graduate from the University of Dhaka (CGPA 3.69/4.00), currently a **Research Associate at BIRDI** working on LLM-guided RTL (Verilog/SystemVerilog) code generation, and a former **Software QA Intern at Samsung R&D Institute Bangladesh**. Experienced in building verified hardware-design datasets, training and evaluating LLMs for code generation, software quality assurance, and full-stack development, with strong teaching and leadership experience.

EDUCATION

University of Dhaka — Institute of Information Technology (IIT) 2022 – Feb 2026
B.Sc. in Software Engineering | CGPA: 3.69 / 4.00

Netrakona Govt. College 2018 – 2020
Higher Secondary Certificate (HSC) | GPA: 5.00 / 5.00

WORK EXPERIENCE

Bangladesh Industry of Research, Development and Innovation (BIRDI) May 2026 – Present
Research Associate — LLM-Guided RTL Code Generation Dhaka, Bangladesh

- Conduct research on LLM-guided RTL code generation covering dataset preparation, model training, and benchmark evaluation, with a focus on improving functional accuracy of generated Verilog.
- Built **CompositeRTL**, a synthetic multi-behavior extension of VerilogEval: generated 6,378 composite spec-RTL-testbench triples, all functionally verified through an automated Icarus Verilog compile-and-simulate loop with LLM-driven repair.
- Fine-tuned Qwen2.5-Coder-7B-Instruct with QLoRA (4-bit NF4, TRL SFTTrainer, PEFT) on a single 16 GB GPU; built config-driven pipelines for training, resumable checkpointing, adapter merging, and inference.
- Diagnosed a post-fine-tuning accuracy regression by compile-checking the SFT corpus — found ~43% of OpenRTLSet samples non-compiling — and added a compile-validation cleaning stage and leakage-free, problem-level train/test splits.
- Developed VerilogEval and RTLLM v2 evaluation harnesses with unbiased pass@k; benchmarked Qwen2.5-Coder (7B/14B), DeepSeek-Coder-6.7B, Qwen3.5-9B, and GPT-series models across difficulty tiers.
- Co-authoring a research paper on the CompositeRTL dataset (in preparation).

Samsung R&D Institute Bangladesh Mar 2025 – Sep 2025
Intern, Quality Innovation Group Dhaka, Bangladesh

- Performed Software Quality Assurance (manual and automated testing); developed test cases and test plans compliant with EAA standards.
- Executed tests, analyzed results, and validated defects using industry automation frameworks and QA methodologies.
- Recognized by the Managing Director for resolving Voice of Customer (VOC) issues on Samsung Notes data-corruption testing.

Institute of Information Technology (IIT), University of Dhaka Apr 2026 – Present
Short Course Instructor (Part-time) Dhaka, Bangladesh

- Teach short professional courses on Python programming, Microsoft Office, and practical AI tools to students and working professionals.
- Design course materials, hands-on exercises, and assessments aligned with industry needs.

Bangladesh Mathematical Olympiad 2023 – 2025
Academic Team Member

- Designed and validated 15+ original problems for regional rounds, contributing to official national Olympiad problem sets.
- Led a team of problem setters and solvers, coordinating task distribution and timely delivery of competition materials.
- Supported event logistics across multiple Olympiad rounds for 500+ contestants.

IIT Software Engineers' Community, University of Dhaka 2024 – 2025
Organizing Secretary

- Planned and coordinated events and knowledge-sharing activities to foster collaboration among members.
- Managed logistics, communications, and event execution to strengthen community engagement.

Dhaka University IT Society 2024 – 2025
Trainer

- Delivered hands-on training on programming, Microsoft Excel, and PowerPoint to 500+ students; developed structured training materials.

Big Bang Academy 2023 – 2024
Mathematics Instructor

- Taught 1,000+ students through interactive, real-time problem-solving sessions.

PROJECTS

AutoDroid — LLM-Guided Automated Android GUI Testing | [GitHub](#)

Python, Android SDK, ADB, uiautomator2, LLMs, NetworkX

- Extended DroidBot with LLM guidance to increase automated GUI test coverage on real Android apps.
- Designed a hybrid strategy — autonomous exploration for fast initial coverage plus on-demand LLM guidance for unexplored features — minimizing costly LLM queries.

The Professional Accountants' Society — Production Web Platform (UK) | [Live](#) | [GitHub](#)

Next.js, TypeScript, Tailwind CSS, MongoDB, Node.js, AWS S3, Vercel

- Built and deployed a full-stack, SEO-friendly platform with server-side rendering; mobile-first UI reduced page load time by up to 35%.
- Implemented cloud deployment and storage (Vercel + AWS S3) achieving 99.9% uptime in production.

BCS Exam Management System | [Live](#) | [GitHub](#)

Next.js 14, TypeScript, Tailwind CSS, FastAPI, PostgreSQL

- Built an examination platform with auto-grading, bulk question upload, and real-time analytics serving 1,000+ students with <500 ms API response time.
- Designed role-based REST APIs (Admin, Moderator, Student, Anonymous) and a normalized PostgreSQL schema (12+ tables) for concurrent exam sessions.

LearnPhysics — Interactive Physics Learning Application | [GitHub](#)

C++, graphics.h

- Built interactive simulations of projectile motion, vectors, and momentum with adjustable parameters and real-time visualization.

TECHNICAL SKILLS

- **LLM & ML:** PyTorch, Hugging Face (Transformers, TRL, PEFT), QLoRA/LoRA, bitsandbytes, Unsloth, TensorFlow, Scikit-learn, NumPy, Pandas, OpenAI API, AWS Bedrock, OpenRouter
- **Hardware Design:** Verilog, SystemVerilog, Icarus Verilog, VerilogEval, RTLLM, testbench development
- **Languages:** Python, C, C++, Java, JavaScript, TypeScript
- **Web:** React.js, Next.js, Node.js, Express.js, FastAPI, Spring Boot, Tailwind CSS, REST APIs
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite
- **Testing & DevOps:** Selenium, Postman, unit/integration/E2E testing, EAA testing, Docker, GitHub Actions, CI/CD, AWS, Vercel, Linux/WSL, Git
- **Professional:** Research & analytical thinking, teaching & mentorship, leadership, communication, project planning

ACHIEVEMENTS

- Recognition gift from the Managing Director, Samsung R&D Institute Bangladesh, for outstanding project contributions.
- Finalist, Project Showcasing, DU ITVerse 2.0 (2024).
- Honorable Mention, ICPC Asia Dhaka Regional Site Online Preliminary Contest (2022).
- Solved 300+ competitive programming problems on [Codeforces](#) and [LeetCode](#).
- “Shera Pathok” award, Bishwa Sahitya Kendra, three consecutive years (2015–2017).

REFERENCES

Dr. Mohammad Shoyaib, Professor

IIT, University of Dhaka
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Dr. Ahmedul Kabir, Associate Professor

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